

Supplementary Material

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1 Implementation Details

All experiments are conducted on an NVIDIA A100 80GB PCIe GPU, using 12 CPU cores with 7.5 GB memory each. The software environment includes CUDA 11.8, PyTorch 2.6.0, and Hugging Face Transformers 4.49.0. To ensure stability and reproducibility, we initialize random states using multiple seeds for Python, NumPy, PyTorch, and CUDA.

Due to the differing nature of encoder-only and decoder-only architectures, we adopt tailored fine-tuning strategies. Encoder-only models use a custom training loop with gradient accumulation for better training control. Decoder-only models (LLMs) are fine-tuned using HuggingFace’s SFTTrainer, with 4-bit quantization via BitsAndBytesConfig for memory efficiency, and LoRA-based parameter-efficient fine-tuning (PEFT) to reduce training overhead while maintaining performance. Fine-tuning results for encoder-only models are averaged over 5 random seeds [42, 123, 456, 789, 1024]; decoder-only models are averaged over 3 seeds [42, 123, 456] due to computational constraints. We also evaluate zero-shot learning by designing a baseline prompt that relies on the model’s pre-trained knowledge without additional training, and few-shot in-context learning by including three task-specific examples within the input prompt. The three examples are selected from the training set using class-balanced sampling (one verifiable, two non-verifiable) with a fixed random seed (42), ensuring deterministic and reproducible example selection across all experiments. The prompt templates are shown in Table 4.

2 Parameter Configurations

For encoder-only models, hyperparameters are optimized on the CT22 development set using the Optuna framework [1] with Bayesian optimization via the Tree-structured Parzen Estimator (TPE) sampler. Each configuration undergoes 20 independent trials, with a MedianPruner strategy to terminate underperforming trials early. Baseline and ContextClaim configurations are optimized independently to ensure a fair comparison. The resulting hyperparameters are applied to both CT22 and PoliClaim without further dataset-specific tuning, so the PoliClaim results reflect cross-dataset transfer of hyperparameter choices. The optimized configurations for the ContextClaim variants are reported in Table 1. For decoder-only models, fine-tuning configurations are selected through empirical tuning to balance computational efficiency and performance, as detailed in Table 2.

In addition to model-specific hyperparameters, the context retrieval component uses fixed parameters throughout all experiments: $p = 5$ candidate extracts per query, relevance score weights $\alpha = 0.8$ and $\beta = 0.2$, and a filtering threshold $\theta = 0.5$ for MISC (and DISEASE) entities. These values were fixed throughout all experiments based on the design rationale and none of these values were tuned on any development set.

Table 1: Hyperparameter configurations for encoder-only ContextClaim models.

Model	Param.	CC-G4o	CC-M
BERT	Epochs	8	8
	Batch	20	20
	Dropout	0.19	0.19
	Warmup	0.14	0.14
	LR	2.5E-05	2.5E-05
RoBERTa	Epochs	8	12
	Batch	32	16
	Dropout	0.23	0.15
	Warmup	0.15	0.15
	LR	3E-05	7.8E-06

Table 2: Decoder-only model hyperparameters.

Parameter	Llama3	Mistral
Epochs	3	3
Batch size	16	8
Warmup ratio	0.1	0.05
Learning rate	1.00E-5	3.00E-4
Optimizer	adamw	paged_adamw*
Grad. accum.	2	4
LoRA r/α	64/16	64/16
LoRA dropout	0.1	0.1
Target modules	q,v,o	q,v

* 8-bit quantization; q,v,o: q_proj, v_proj, o_proj; weight decay: 0.001; max grad. norm: 1.0; scheduler: cosine w/ restarts.

Table 3: Prompt for context summarization.

Summarization Prompt
You are a helpful assistant. Provide a factual summarization around 150 words. Input claim: "{claim_text}" Relevant Context: {extracts_from_retrieved_wiki_pages} Generate a concise, objective summary to the provided claim based ONLY on the provided context.

3 Prompts

This section provides the complete prompt templates used in our experiments. Table 3 shows the prompt for generating context summaries from retrieved Wikipedia content, and Table 4 presents our prompt template. The baseline prompt uses the same structure but omits the “Additional information” component.

Table 4: Prompt template for verifiable claim detection. The baseline prompt excludes the highlighted components.**Prompt Template**

Instruction:

Determine if the input text contains verifiable claims.
The input text contains verifiable claims if it makes specific factual statements that can be checked against evidence.

Additional information may help clarify what the claim refers to, but base your decision primarily on whether the claim makes specific factual statements.

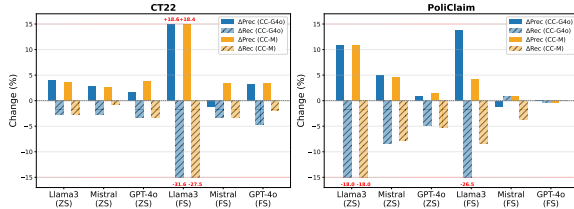
If the input text contains claims that can be verified, respond "Yes". Otherwise, respond "No". Note: When in doubt, choose "Yes".

In the end, respond only with 'Yes' for verifiable claims or 'No' for unverifiable claims.

Input text: {claim_text}

Additional information: {context_summary}

Response: {Yes/No}

**Figure 1: Effect of Removing Prompt Bias: Precision vs Recall Trade-off.****Table 5: Effect of Removing Prompt Bias ("When in doubt, choose Yes"). Δ = Ablation – Original. Bold: $|\Delta| > 3.5$.**

Set.	Model	CC-G4o			CC-M		
		Δ Prec	Δ Rec	Δ F1	Δ Prec	Δ Rec	Δ F1
CT22							
ZS	Llama3	+3.95	-2.68	+2.13	+3.70	-2.68	+1.94
	Mistral	+2.88	-2.69	+1.20	+2.73	-0.67	+1.76
	GPT-4o	+1.60	-3.36	-0.33	+3.75	-3.36	+0.92
FS	Llama3	+18.57	-31.55	-4.63	+18.44	-27.52	-1.71
	Mistral	-1.18	-3.35	-2.03	+3.42	-3.36	+0.98
	GPT-4o	+3.31	-4.69	-1.23	+3.38	-2.01	+0.45
PoliClaim							
ZS	Llama3	+10.88	-18.04	-2.36	+10.81	-18.04	-2.31
	Mistral	+5.01	-8.44	-1.46	+4.59	-7.87	-1.41
	GPT-4o	+0.97	-4.80	-3.17	+1.41	-5.19	-3.24
FS	Llama3	+13.78	-26.49	-7.37	+4.18	-8.45	-2.18
	Mistral	-1.11	+0.77	-0.31	+0.87	-3.64	-1.34
	GPT-4o	+0.00	-0.38	-0.36	-0.42	+0.00	-0.08

3.1 Prompt Bias Analysis

The precision-recall trade-offs discussed in Section 5.1 of the main paper reveal an interesting pattern. Decoder-only models under zero-shot and few-shot settings often achieve near-perfect recall, but precision suffers considerably as a result. We suspect part of this imbalance comes from a specific directive in the original prompt (Table 4), namely the instruction "Note: When in doubt, choose 'Yes'", which tells models to lean toward predicting verifiable when they are uncertain. To examine whether this instruction is actually driving the imbalance, we conducted a prompt ablation study. We removed the directive and kept everything else unchanged, then re-ran all zero-shot and few-shot experiments across both datasets to see how precision and recall would respond. Figure 1 reports the changes in precision and recall after the ablation, and Table 5 provides the full breakdown. Removing the directive shifts predictions in a consistent direction across nearly all configurations, with precision going up and recall going down in most cases. This pattern supports our hypothesis that the "when in doubt" instruction is a direct contributor to the false positive bias discussed in the main paper. The directive nudges the model toward predicting verifiable on ambiguous samples, which boosts recall but hurts precision. The size of this effect, however, varies quite a bit depending on the dataset and the model. On CT22, the changes are generally moderate, and most configurations stay within a narrow range. Llama3 under few-shot stands out as an exception, where removing the directive produces a substantial precision gain together with a noticeable drop in recall. Another telling case is Llama3 zero-shot on CT22. With the original prompt, the model essentially predicts verifiable for every sample, reaching perfect recall at the cost of much lower precision. Once the directive is removed, the model arrives at a more balanced operating point, recovering reasonable precision while keeping recall high. On PoliClaim, the effect is more pronounced and shows up across more configurations. Llama3 and Mistral both shift substantially in both zero-shot and few-shot settings. GPT-4o, by contrast, remains fairly stable on both datasets, with only minor changes overall. This suggests that a sufficiently capable model tends to resolve ambiguity using its own parametric knowledge instead of relying on the prompt directive for guidance. It is also worth noting that although precision improves in many configurations, the ablation actually leads to lower F1 in most cases, which indicates that the directive, despite introducing bias, helps the model capture genuine verifiability signals that would otherwise be missed. Taken together, these findings suggest that the "when in doubt" instruction acts as a kind of recall safety net, and is especially valuable for weaker models operating without demonstrations. Future work could look into alternative formulations that preserve recall without amplifying false positives.

4 Context Summary Quality Assessment

To evaluate the quality of generated context summaries, we employed human annotators to assess 160 claim-summary pairs (40 samples $\times$ 2 datasets $\times$ 2 LLM-based generators) across three dimensions using a 3-point Likert scale (1=Poor, 2=Acceptable, 3=Good), as described in Section 6.2. Since both datasets are publicly available, the evaluation posed minimal privacy or ethical concerns. Annotators were trained following the guidelines presented below

to assess whether summaries provide useful context for determining claim verifiability, rather than assessing the truthfulness of the claims themselves.

Relevance. How relevant is the summary to the claim being assessed?

- Poor (1): Content is off-topic, unrelated to the claim, or discusses completely different or not mentioned entities/events.
- Acceptable (2): Provides some relevant background but includes unnecessary tangential information or misses some related aspects.
- Good (3): Content directly relates to the entities, events, or concepts mentioned in the claim; stays on topic.

Signal Clarity. Does the summary provide clear signals for determining verifiability, such as whether entities are real or fictional, events are documentable or hypothetical, statements are objective or subjective?

- Poor (1): Provides no clear signals about verifiability; leaves it unclear whether the claim can be fact-checked, e.g., only gives general background without addressing existence or documentability of specific entities or events.
- Acceptable (2): Provides some verifiability signals but they are implicit, incomplete, or ambiguous, e.g., confirms an

entity exists but does not clarify whether the specific event is documentable.

- Good (3): Explicitly provides clear signals that help determine verifiability, e.g., confirms entities are real persons/organizations, indicates whether events are recorded, and clarifies whether statements are objective or subjective.

Usefulness. How confident are you in determining claim verifiability after reading the summary?

- Poor (1): The summary does not help at all; you feel equally uncertain about the claim's verifiability as before reading it.
- Acceptable (2): The summary provides some help but leaves significant uncertainty; you would need additional information to make a confident verifiability judgment.
- Good (3): The summary significantly helps you determine whether the claim is verifiable or unverifiable; you feel confident in making this assessment based on the provided context.

References

- [1] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. 2019. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2623–2631.